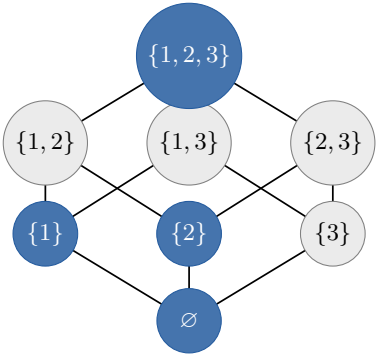
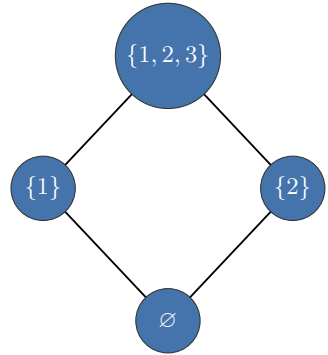
(a) Ambient Boolean lattice $\mathcal{P}(\{1,2,3\})$



$B = \{\varnothing, \{1\}, \{2\}, \{1,2,3\}\}$

Ambient join:
$\{1\} \vee_L \{2\} = \{1,2\} \notin B$

(b) Induced order on B



$\{1\} \vee_B \{2\} = \{1,2,3\}, \quad \{1\} \wedge_B \{2\} = \varnothing$

B is an intrinsic lattice,
but not an ambient sublattice